

SAFETY DATA SHEET (GHS)

Diisopropanolamine (DIPA) — PLACEHOLDER for 'Diamine — TBC' inhibitor stream | CAS: 110-97-4 | Formula: C6H15NO2

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

PLACEHOLDER SDS — CHEMICAL IDENTITY NOT YET CONFIRMED. The HAZOP PSI brief lists this process item as 'Diamine — TBC (To Be Confirmed)', used in conjunction with TBC (tert-butylcatechol, the polymerisation inhibitor for AMS/styrenic streams). Per user instruction, Diisopropanolamine (DIPA, CAS 110-97-4) has been used as a PLACEHOLDER substance to demonstrate the SDS format and provide a representative amine hazard profile pending confirmation of the actual diamine product identity used at PPCL. DO NOT rely on this SDS for the actual process chemical until the true identity/CAS number of the diamine additive is confirmed and a corresponding SDS is generated.

DATA SOURCES CONSULTED

- ChemicalBook — Diisopropanolamine, CAS 110-97-4, <https://www.chemicalbook.com/>
- TCI America — Product D0924, Diisopropanolamine (CAS 110-97-4), <https://www.tcichemicals.com/>
- chemBlink — Diisopropanolamine SDS, CAS 110-97-4, <https://www.chemblink.com/>
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Diisopropanolamine (DIPA) — PLACEHOLDER, see note above

CAS Number: 110-97-4 (PLACEHOLDER — actual diamine CAS TBC)

Chemical formula: C6H15NO2 (DIPA)

Molecular weight: 133.19 g/mol (DIPA)

Synonyms: Bis(2-hydroxypropyl)amine; 1,1'-iminodi-2-propanol

Process role: PLACEHOLDER for the 'Diamine — TBC' additive/inhibitor stream referenced in the HAZOP PSI brief, used in conjunction with TBC (tert-butylcatechol) for polymerisation inhibition of AMS/styrenic process streams. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification (DIPA, placeholder): Eye Irrit. 2A (H319)

Signal word: Warning

Hazard statements (H-codes): H319 (Causes serious eye irritation)

Precautionary statements (P-codes): P264, P280, P305+P351+P338, P337+P313, P501

GHS pictograms: GHS07 (Exclamation mark)

Other hazards: PLACEHOLDER CAVEAT: this hazard profile is for DIPA only and may NOT represent the actual diamine product used at PPCL. DIPA is described in available sources as a corrosive, hygroscopic solid; combustible when exposed to heat or flame; dust explosion is possible if handled in powder/granular form mixed with air (relevant if the actual diamine additive is also a solid amine handled as powder). Amines of this general type can react with TBC (a phenolic inhibitor) and other process chemicals — compatibility of the actual diamine additive with TBC and with peroxide-bearing streams (CHP) should be specifically assessed once the actual identity is confirmed.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Substance (placeholder): Diisopropanolamine, CAS 110-97-4 — PLACEHOLDER ONLY. Actual diamine additive composition/concentration TBC.

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove to fresh air. If dust is inhaled (solid handling), seek medical attention if respiratory irritation persists.

Skin contact: Wash with plenty of water. Remove contaminated clothing. DIPA is described as corrosive in some sources — obtain medical attention if irritation/burns occur.

Eye contact: Flush immediately with copious water for at least 15 minutes, holding eyelids open. Obtain medical attention (H319).

Ingestion: Rinse mouth with water. Do not induce vomiting. Obtain medical attention immediately.

Notes to physician: Treat as alkaline amine exposure; assess for corrosive/irritant effects on mucous membranes

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Dry chemical, CO₂, water spray, alcohol-resistant foam

Unsuitable extinguishing media: None specific identified

Specific hazards: DIPA is combustible when exposed to heat or flame; if handled as a dry powder/granular solid, dust explosion is possible when mixed with air — relevant if the actual diamine additive is also handled as a solid

Special protective equipment for firefighters: Self-contained breathing apparatus (SCBA) and full protective clothing

Hazardous combustion products: Carbon monoxide, carbon dioxide, nitrogen oxides (NO_x) — typical of amine combustion

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Avoid skin/eye contact and dust inhalation; wear appropriate PPE (Section 8)

Environmental precautions: Avoid release to drains/watercourses; amines can affect receiving water pH and aquatic life

Containment and cleanup: Sweep up solid material (if applicable) or absorb liquid/molten material with inert absorbent; collect into closed, labelled containers for disposal

Reference to other sections: See Section 7 (Handling/Storage)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Avoid dust generation if handled as solid (dust explosion possibility); avoid skin/eye contact; ensure adequate ventilation

Storage conditions: Store in tightly closed containers in a cool, dry area; DIPA is hygroscopic — protect from moisture ingress

Temperature limits: DIPA melts at approximately 42-45°C — confirm storage temperature requirements (solid vs. molten handling) once actual diamine identity is confirmed

Pressure: Atmospheric storage

Segregation requirements: Segregate from strong acids (exothermic neutralisation), strong oxidisers, and — pending confirmation of compatibility — from CHP/peroxide-bearing streams (CAS 80-15-9) and TBC until co-formulation compatibility is verified

Special requirements: PRIORITY ACTION: confirm actual diamine product identity and CAS number, then replace this placeholder SDS with the correct substance-specific SDS before this document is used for PSI/HAZOP purposes

Specific end-use recommendations: Once confirmed, assess compatibility of the actual diamine additive with TBC inhibitor and with AMS/CHP streams (see AMS SDS, CAS 98-83-9, Section 7, regarding inhibitor management)

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): No specific ACGIH TLV or OSHA PEL established for DIPA; confirm for actual diamine additive once identified

TWA: Not established (DIPA)

STEL: Not established

Ceiling: Not established

Skin notation: Not assigned

IARC/NTP/OSHA carcinogen status: Not classified as a carcinogen (DIPA)

Engineering controls: General ventilation; local exhaust/dust control if solid handling is involved

PPE — Respiratory: Dust mask/respirator if handling solid material with dust generation

PPE — Gloves: Chemical-resistant gloves (e.g., nitrile)
PPE — Eyes: Chemical splash goggles (H319)
PPE — Body: Standard work clothing; chemical-resistant apron for bulk transfer
Biological monitoring: Not applicable

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: White to off-white solid (flakes/granules) or viscous liquid above melting point (DIPA placeholder)
Odour: Mild amine odour
Odour threshold: Not established
pH: Strongly alkaline in aqueous solution (amine)
Melting/Freezing point: ~42-45°C (DIPA)
Boiling point (at 1 atm): ~248-249°C (DIPA, with decomposition possible near boiling point)
Flash point: Not well established for DIPA from available sources — confirm; amino alcohols of this type typically have flash points in the 150-180°C range
Evaporation rate: Very low (low volatility solid/high-boiling liquid)
Flammability (solid/gas): Combustible solid when exposed to heat/flame; dust explosion possible if handled as fine powder
Flammability limits in air: Not established for DIPA; if actual diamine is handled as powder, dust explosibility (Kst/Pmax) should be characterised
Vapour pressure: Very low at ambient temperature
Vapour density (air = 1): ~4.6 (estimated from molecular weight, DIPA)
Relative density (water = 1): ~0.989 (DIPA, molten)
Solubility (water): Miscible/highly soluble (DIPA)
Partition coefficient (log Kow): ~0.1-0.3 (estimated, DIPA — hydrophilic)
Auto-ignition temperature: Not established
Decomposition temperature: DIPA may decompose near its boiling point (~248-249°C); confirm for actual diamine additive
Viscosity: Not established (DIPA is solid at room temperature)
Explosive properties: Not classified as explosive in liquid/dissolved form; dust explosion possibility if handled as fine powder
Oxidising properties: Not an oxidiser
Molecular weight: 133.19 g/mol (DIPA)

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: DIPA is stable under normal conditions; reacts with strong acids (exothermic neutralisation, forming amine salts) and may react with strong oxidisers
Chemical stability: Stable under recommended storage conditions
Possibility of hazardous reactions: Exothermic reaction with strong acids; potential incompatibility with peroxides/CHP (amines can interact with organic peroxides — this should be specifically assessed for the actual diamine additive given its intended co-location with CHP-containing process streams)
Conditions to avoid: Contact with strong acids, strong oxidisers, and — pending confirmation — peroxide-bearing streams
Incompatible materials: Strong acids, strong oxidisers, possibly organic peroxides (CHP) — CONFIRM for actual diamine additive
Hazardous decomposition products: Nitrogen oxides (NOx), carbon monoxide, carbon dioxide on combustion/thermal decomposition

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Skin/eye contact, inhalation (dust), ingestion
Acute toxicity — Oral LD50: ~5,900-7,400 mg/kg, rat (DIPA — low acute oral toxicity)
Acute toxicity — Dermal LD50: Not well established (DIPA)
Acute toxicity — Inhalation LC50: Not established

Skin corrosion/irritation: Some sources describe DIPA as corrosive to skin; confirm classification for actual diamine additive

Serious eye damage/irritation: Causes serious eye irritation (H319, DIPA)

Respiratory/skin sensitisation: Not classified as a sensitiser (DIPA); confirm for actual diamine additive

Germ cell mutagenicity: Not classified (DIPA)

Carcinogenicity: Not classified as a carcinogen (DIPA)

Reproductive toxicity: Not classified (DIPA)

STOT (single exposure): Not classified (DIPA)

STOT (repeated exposure): Not classified (DIPA)

Aspiration hazard: Not applicable (solid, DIPA)

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC50/EC50): Not well established for DIPA from available sources; amino alcohols generally show low-to-moderate aquatic toxicity

Persistence and degradability: DIPA is expected to be biodegradable

Bioaccumulative potential: Low (hydrophilic, low log Kow)

Mobility in soil: High mobility (highly water soluble)

Other adverse effects: None specifically identified for DIPA

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Dispose of via licensed hazardous waste incineration or treatment facility in accordance with local regulations; do not discharge to drains without neutralisation/treatment

Contaminated packaging: Dispose of as hazardous waste or, if uncontaminated/empty, per local regulations after appropriate purging

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535 (verify applicability); local industrial waste regulations

SECTION 14 — TRANSPORT INFORMATION

UN Number: Not regulated as a dangerous good for transport based on available DIPA data (confirm for actual diamine additive)

Proper shipping name: Not applicable / Not regulated (DIPA placeholder)

Transport hazard class(es): Not classified as a transport hazard (DIPA placeholder)

Packing group: Not applicable

Environmental hazards (marine pollutant): Not classified as a marine pollutant (DIPA placeholder)

Special precautions for user: None specific identified for DIPA; CONFIRM for actual diamine additive

Bulk transport (MARPOL/IBC): Not established — confirm for actual diamine additive

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below — NOTE: these regulatory notes pertain to DIPA (CAS 110-97-4) only and should be re-verified once the actual diamine additive identity is confirmed

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0 — PLACEHOLDER

Key changes from previous version: Initial issue — PLACEHOLDER SDS using Diisopropanolamine (CAS 110-97-4) compiled for HAZOP PSI in place of the unconfirmed 'Diamine — TBC' process additive, per explicit user instruction.

ACTION REQUIRED: PPCL process engineering/HSE to confirm actual diamine product identity and CAS number; this placeholder SDS must be replaced with a substance-specific SDS before reliance on this document for PSI/HAZOP purposes.

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document (DIPA data only — PLACEHOLDER)

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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